ML Viva Questions & Answers

Unit I – Introduction to Machine Learning

Q1. What is Machine Learning?

Ans. Machine Learning (ML) is a branch of AI that allows computers to learn patterns from data and make predictions without explicitly programming every rule.

Q2. What is the difference between ML and traditional programming?

Ans. Traditional programming uses predefined rules to generate outputs. ML learns patterns from data and uses them to make predictions.

Q3. What is the difference between AI, ML, and Data Science?

Ans. AI focuses on making machines intelligent. ML enables machines to learn from data. Data Science extracts useful information from data using statistics and computing.

Q4. What are the types of Machine Learning?

Ans. There are four main types: Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning.

Q5. What is Supervised Learning?

Ans. Supervised Learning uses labeled data containing both input features and correct outputs. Examples: Regression and Classification.

Q6. What is Unsupervised Learning?

Ans. Unsupervised Learning finds hidden patterns in unlabeled data. Example: K-Means Clustering.

Q7. What is Semi-Supervised Learning?

Ans. Semi-Supervised Learning combines a small amount of labeled data with a large amount of unlabeled data.

Q8. What is Reinforcement Learning?

Ans. Reinforcement Learning trains an agent to make decisions through interactions with an environment using rewards and penalties.

Q9. What is the difference between Classification and Regression?

Ans. Classification predicts categories, such as Spam or Not Spam. Regression predicts continuous numerical values, such as house prices.

Q10. What is a Parametric Model?

Ans. A Parametric Model has a fixed number of parameters regardless of training data size. Example: Linear Regression.

Q11. What is a Non-Parametric Model?

Ans. A Non-Parametric Model has flexibility that can grow with the training data. Example: KNN.

Q12. What are Geometric, Probabilistic, and Logical Models?

Ans. Geometric models use distances or boundaries, Probabilistic models use probabilities, and Logical models use rules and conditions.

Q13. What are Grouping and Grading Models?

Ans. Grouping models organize similar data into clusters. Grading models assign scores or ranks to data.

Q14. What is Learnability in ML?

Ans. Learnability means whether a model can learn useful patterns from training data and perform well on unseen data.

Q15. What are common data formats in ML?

Ans. Structured data like tables, semi-structured data like JSON, and unstructured data like text and images.

Unit II – Feature Engineering

Q16. What is Feature Engineering?

Ans. Feature Engineering is the process of creating, selecting, and transforming input features to improve model performance.

Q17. What is Data Preprocessing?

Ans. Data Preprocessing cleans and prepares raw data before model training. It includes handling missing values, scaling, and removing invalid data.

Q18. What is a Feature and a Target Variable?

Ans. A feature is an input used for prediction. A target variable is the output that the model tries to predict.

Q19. What is Normalization?

Ans. Normalization commonly scales numerical values between 0 and 1 using Min-Max Scaling.

Formula: $x'=\frac{x-x_{\min}}{x_{\max}-x_{\min}}$

Q20. What is Standardization?

Ans. Standardization transforms values to have mean 0 and standard deviation 1.

Formula: $z=\frac{x-\mu}{\sigma}$

Q21. What is the difference between Normalization and Standardization?

Ans. Normalization usually scales data to a fixed range. Standardization centers data around zero and scales by standard deviation.

Q22. How do you handle missing values?

Ans. Missing values can be handled by removing records or filling them using mean, median, or mode, depending on the data.

Q23. What is an Outlier?

Ans. An outlier is a data point significantly different from other observations. It can be detected using IQR, Z-score, or box plots.

Q24. What is Dimensionality Reduction?

Ans. Dimensionality Reduction reduces the number of features while preserving important information.

Q25. What is PCA?

Ans. Principal Component Analysis (PCA) reduces dimensionality by converting correlated features into new uncorrelated components that capture maximum variance.

Q26. What is Kernel PCA?

Ans. Kernel PCA is a nonlinear dimensionality reduction technique that uses kernel functions to identify complex patterns.

Q27. What is Local Binary Pattern (LBP)?

Ans. LBP is an image texture feature extraction method that compares neighboring pixel values to form binary patterns.

Q28. What is the difference between Feature Selection and Feature Extraction?

Ans. Feature Selection chooses useful original features. Feature Extraction creates new features from existing data.

Q29. What is Sequential Forward Selection?

Ans. Sequential Forward Selection (SFS) starts with no features and adds the most useful feature one by one.

Q30. What is Sequential Backward Selection?

Ans. Sequential Backward Selection (SBS) starts with all features and removes the least useful features one by one.

Q31. What is Statistical Feature Engineering?

Ans. It creates new features using statistical values such as count, length, mean, median, and mode.

Q32. What is Multidimensional Scaling (MDS)?

Ans. MDS reduces dimensions while trying to preserve the distances between data points.

Q33. What is Matrix Factorization?

Ans. Matrix Factorization represents a matrix as the product of smaller matrices to discover patterns or reduce dimensions.

Unit III – Supervised Learning: Regression

Q34. What is Linear Regression?

Ans. Linear Regression predicts continuous numerical values by finding a linear relationship between input features and output.

Formula: $y = mx + c$.

Q35. What is Bias in ML?

Ans. Bias is the error caused by overly simple assumptions in a model. High bias can cause underfitting.

Q36. What is Variance in ML?

Ans. Variance describes how much model predictions change with different training datasets. High variance can cause overfitting.

Q37. What is Overfitting?

Ans. Overfitting happens when a model learns training data too closely, including noise, and performs poorly on new data.

Q38. What is Underfitting?

Ans. Underfitting happens when a model is too simple to capture important patterns in the data.

Q39. What is Generalization?

Ans. Generalization is the ability of a trained model to make accurate predictions on unseen data.

Q40. What is the Bias-Variance Tradeoff?

Ans. It is the balance between high bias (underfitting) and high variance (overfitting) to improve prediction performance.

Q41. What is Ridge Regression?

Ans. Ridge Regression uses L2 regularization to penalize large coefficients and reduce overfitting.

Q42. What is Lasso Regression?

Ans. Lasso Regression uses L1 regularization and can reduce some feature coefficients to zero.

Q43. What is the difference between Ridge and Lasso?

Ans. Ridge (L2) shrinks coefficients toward zero. Lasso (L1) can make some coefficients exactly zero, performing feature selection.

Q44. What is Gradient Descent?

Ans. Gradient Descent is an optimization algorithm that repeatedly updates parameters in the negative gradient direction to minimize a loss function.

Q45. What is the Gradient Descent Formula?

Ans. $\theta_{new} = \theta_{old} - \alpha \times \text{gradient}$, where α is the learning rate.

Q46. What is Learning Rate?

Ans. Learning Rate controls the step size during Gradient Descent. A very high value may cause divergence, while a very low value slows training.

Q47. What is MAE?

Ans. Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values. Lower MAE is better.

Q48. What is RMSE?

Ans. Root Mean Squared Error (RMSE) is the square root of the average squared prediction error. It penalizes large errors more strongly.

Q49. What is R^2 Score?

Ans. R^2 Score measures how well predictions explain variation in the target compared with predicting the target mean. Closer to 1 is better.

Unit IV – Supervised Learning: Classification

Q50. What is K-Nearest Neighbors (KNN)?

Ans. KNN is a supervised learning algorithm that classifies a data point using the majority class among its K nearest neighbors.

Q51. Explain the working of KNN.

Ans.

1. Choose K.
2. Calculate distances.
3. Select K nearest neighbors.
4. Assign the majority class.

Q52. What is the Euclidean Distance Formula?

Ans. Euclidean Distance calculates the straight-line distance between two points.

Formula: $d=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$

Q53. What happens when K is too small or too large?

Ans. Small K can cause overfitting because it is sensitive to noise. Large K can cause underfitting by oversmoothing class boundaries.

Q54. What is Support Vector Machine (SVM)?

Ans. SVM is a supervised algorithm that finds a decision boundary with the maximum possible margin between classes.

Q55. What are Support Vectors?

Ans. Support Vectors are training points that determine the position of the SVM decision boundary.

Q56. What is the Kernel Trick in SVM?

Ans. The Kernel Trick allows SVM to create nonlinear decision boundaries without explicitly calculating high-dimensional feature mappings.

Q57. What is Ensemble Learning?

Ans. Ensemble Learning combines predictions from multiple models to improve accuracy and stability.

Q58. What is Bagging?

Ans. Bagging trains multiple models on different bootstrap samples and combines their predictions to reduce variance.

Q59. What is Boosting?

Ans. Boosting trains models sequentially, with later models focusing on improving earlier predictions.

Q60. What is Random Forest?

Ans. Random Forest combines multiple decision trees trained with randomness to improve prediction accuracy and reduce overfitting.

Q61. What is AdaBoost?

Ans. AdaBoost is a boosting algorithm that increases the importance of previously misclassified samples.

Q62. What is the difference between Bagging and Boosting?

Ans. Bagging trains models largely independently and combines predictions. Boosting trains models sequentially to correct previous errors.

Q63. What is Binary and Multiclass Classification?

Ans. Binary Classification predicts two classes, like Spam/Not Spam. Multiclass Classification predicts three or more classes.

Q64. What is Balanced and Imbalanced Classification?

Ans. Balanced data has roughly equal samples per class. Imbalanced data has unequal class distributions.

Q65. What is One-vs-One and One-vs-All Classification?

Ans. One-vs-One trains a classifier for each pair of classes. One-vs-All trains one classifier for each class against the rest.

Q66. What is a Confusion Matrix?

Ans. A Confusion Matrix compares actual and predicted classes using True Positive, True Negative, False Positive, and False Negative.

Q67. What is Accuracy?

Ans. Accuracy is the proportion of correctly classified samples.

Formula: $\frac{TP+TN}{TP+TN+FP+FN}$

Q68. What is Precision?

Ans. Precision measures how many predicted positive samples are actually positive.

Formula: $\frac{TP}{TP+FP}$

Q69. What is Recall?

Ans. Recall measures how many actual positive samples are correctly identified.

Formula: $\frac{TP}{TP+FN}$

Q70. What is F1-Score?

Ans. F1-Score is the harmonic mean of precision and recall.

Formula: $\frac{2PR}{P+R}$

Q71. What is Cross-Validation?

Ans. Cross-Validation evaluates a model on different training and validation splits to estimate how well it generalizes.

Q72. What is Micro-Averaging and Macro-Averaging?

Ans. Micro-Averaging combines class-wise counts before calculating metrics. Macro-Averaging calculates metrics for each class and takes their average.

Unit V – Unsupervised Learning

Q73. What is K-Means Clustering?

Ans. K-Means groups data into K clusters by assigning points to their nearest centroid and repeatedly updating centroids.

Q74. Explain K-Means Algorithm.

Ans.

1. Select K initial centroids.
2. Assign points to nearest centroids.
3. Update centroid positions.
4. Repeat until convergence.

Q75. What is K-Medoids?

Ans. K-Medoids is a clustering algorithm that uses actual data points as cluster centers, making it generally more resistant to outliers.

Q76. What is Hierarchical Clustering?

Ans. Hierarchical Clustering creates a tree of nested clusters by repeatedly merging or splitting groups.

Q77. What is a Dendrogram?

Ans. A Dendrogram is a tree diagram showing how clusters are merged or divided.

Q78. What is DBSCAN?

Ans. DBSCAN is a density-based clustering algorithm that groups closely packed points and identifies noise.

Q79. What is Spectral Clustering?

Ans. Spectral Clustering uses eigenvectors of a similarity-based graph matrix to identify clusters.

Q80. What is Isolation Forest?

Ans. Isolation Forest detects outliers by randomly partitioning data. Outliers usually need fewer splits to isolate.

Q81. What is Local Outlier Factor (LOF)?

Ans. LOF identifies outliers by comparing the local density of a point with the densities of nearby points.

Q82. What is the Elbow Method?

Ans. The Elbow Method selects K by plotting WCSS against the number of clusters and looking for a bend in the curve.

Q83. What is WCSS?

Ans. Within-Cluster Sum of Squares (WCSS) measures the total squared distance between points and their cluster centroids.

Q84. What is Intrinsic and Extrinsic Clustering Evaluation?

Ans. Intrinsic Evaluation uses properties of the clustered data, such as silhouette score. Extrinsic Evaluation compares clusters with reference labels.

Unit VI – Introduction to Neural Networks

Q85. What is an Artificial Neural Network (ANN)?

Ans. ANN is a machine learning model consisting of connected artificial neurons that learn patterns from data.

Q86. What is a Single-Layer Neural Network?

Ans. A Single-Layer Neural Network has input connections directly to the output layer without hidden layers.

Q87. What is a Multilayer Perceptron (MLP)?

Ans. MLP is a feedforward neural network containing one or more hidden layers with nonlinear activation functions.

Q88. What is an Activation Function?

Ans. An Activation Function transforms a neuron's output and introduces nonlinearity. Examples: ReLU, Sigmoid, Tanh, and Softmax.

Q89. What is Backpropagation?

Ans. Backpropagation calculates gradients of the loss with respect to network weights using the chain rule, allowing an optimizer to update them.

Q90. What is FLANN?

Ans. Functional Link Artificial Neural Network (FLANN) expands input features using mathematical functions, usually without hidden layers.

Q91. What is an RBF Network?

Ans. Radial Basis Function (RBF) Network uses distance-based activation functions in its hidden layer to learn patterns.

Q92. What is a Recurrent Neural Network (RNN)?

Ans. RNN processes sequential data by passing information from previous time steps to later steps.

Q93. What is a Convolutional Neural Network (CNN)?

Ans. CNN uses convolution filters to extract spatial patterns from images or other grid-like data.

Group B – Most Important Practical Viva Questions

Q94. Which algorithms are used for Uber Fare Prediction?

Ans. Linear Regression and Random Forest Regression. Their performance is compared using R^2, RMSE, and MAE.

Q95. Which algorithms are used for Email Spam Detection?

Ans. KNN and SVM are used for binary classification into Spam and Not Spam.

Q96. Which model is used for Bank Customer Churn Prediction?

Ans. Artificial Neural Network (ANN) is used to predict whether a customer will leave the bank. The model is evaluated using accuracy and confusion matrix.

Q97. What is the minimum of $y = (x + 3)^2$ in Gradient Descent practical?

Ans. The function reaches its global minimum at $x = -3$, where $y = 0$. Gradient Descent starts from $x = 2$.

Q98. Which algorithm is used for Diabetes Prediction?

Ans. K-Nearest Neighbors (KNN) is used. Performance is evaluated using confusion matrix, accuracy, error rate, precision, and recall.

Q99. Which algorithms are used for Sales Data Clustering?

Ans. K-Means or Hierarchical Clustering is used. The Elbow Method helps select the number of clusters for K-Means.

Q100. Why do we split a dataset into Training and Testing Sets?

Ans. Training data is used to teach the model, while testing data evaluates its performance on unseen samples and helps detect overfitting.

Quick Revision – ML Algorithms and Their Uses

Algorithm	Main Use	Type
Linear Regression	Predict continuous values	Supervised
Ridge / Lasso	Regularized regression	Supervised
KNN	Classification	Supervised
SVM	Classification	Supervised
Random Forest	Classification / Regression	Supervised
AdaBoost	Ensemble prediction	Supervised
K-Means	Clustering	Unsupervised
K-Medoids	Clustering	Unsupervised
Hierarchical Clustering	Clustering	Unsupervised
DBSCAN	Density-based clustering	Unsupervised
Isolation Forest	Outlier detection	Unsupervised
ANN / MLP	Classification / Regression	Supervised (commonly)
RNN	Sequential data	Neural network
CNN	Image processing	Neural network